Entropy reduction in mathematical giftedness

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A new measurement for mental performance

This paper deals with the elementary analysis of cognitive processes in mathematical problem solving for better diagnostics. The present experiment was designed to analyse the internal process and to localize the neural substrates involved in solving mathematical tasks, using EEG-coherence. The internal process is revealed by a sequence of microstates. Microstates are defined here as coherence maps which remain stable over time. The difference in performance between gifted and normal subjects in solving mathematical problems is reflected by the difference between the sequential and topographical properties of microstate sequences. Sequential properties were measured by means of entropy reduction. A higher entropy reduction was found in mathematically gifted subjects in contrast to the normal subjects. Topographical properties were measured by means of difference maps of microstates between mathematically gifted and normal subjects. Gifted subjects exhibit a higher coherence left and right parietal and left frontal. This result corroborates the double representation hypothesis.
This new method may be extended to measure mental performance more objectively.

The so-called double-representation hypothesis plays an important role in mathematical (and technical) giftedness. Obviously, particular mathematical problems can be solved by means of computation (e.g. using the equation of Pythagoras) or based on a visual mental imagery resolution strategy (e.g. using the imagery of triangles) (see Fig. 1). Mathematically gifted subjects may be able to activate both the computational and the imagery resolution strategy and to switch from one to the other (Klix 1992). In order to determine the differences in ability between gifted and normal subjects, we used mathematical tasks solvable with both modality strategies: the computation component of numerical mental calculation and the imagery component of figural mental calculation (Heinrich 1997). The strategies differ in the mental effort required.

Gifted subjects exhibit high-level cognitive performance. As an example, one of our mathematically gifted subject solved the following problem: `What is the number of diagonals of a 23-polygon?` in 28 seconds. To decide whether this phenomenon is caused by better cognitive process, an EEG-coherence analysis was used to subdivide the thinking process into a sequence of microstates with sequential and topographical properties. Sequential properties: Cognitive process is reflected by means of entropy reduction, as a kind of concatenation of

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microstates. Topographical properties: EEG-coherence maps were used to determine the activated cortical areas.

Method

Subjects. Twelve right handed normal (16 to 18 years old; mean, 17.5; 6 females) and twelve right handed highly mathematical gifted subjects (16 to 24 years old; mean, 17.9; 3 females) participated in the experiment. The normal subjects were enrolled at an ordinary highschool. The highly mathematical gifted subjects were enrolled at the special mathematical highschool. In order to exclude covariables, further criteria were proved. Firstly, they were tested for their performance on the verbal and non-verbal IQ test (Table 1). The two samples exhibited a significant difference on the non-verbal, but not in the verbal IQ test. Secondly, both samples scored within the normal range in metal rotation tests and memory capacity (Cavanagh-constante) although their visual digit-span differed.

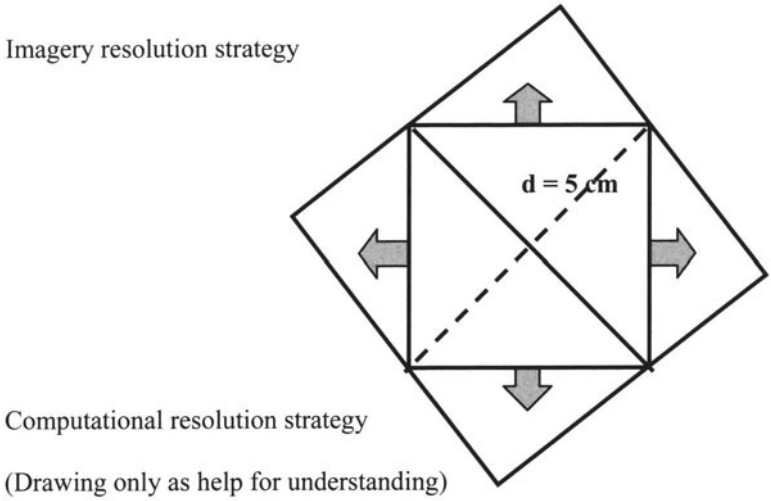
Table 1. Intelligence quotient (IQ), mental rotation (scores), Cavanagh-constant (ms), memory span for digits, working memory capacity and response latencies (sec) for mathematically gifted and normal subjects (SD = standard deviation)

Test–score	p-value		gifted	Normal	Z
IQ nonverbal		mean	124	107	-
2.79	0.004				
		SD	10	10	
IQ verbal		mean	112	105	-
0.997	0.34				
		SD	13	4	
Mental rotation		mean	17	14	-
1.317	0.193				
		SD	3	5	
Cavanagh-constant		mean	295	247	-
0.821	0.438				
		SD	116	77	
emory span for dig		mean	7.36	6.03	-
3.45	0.000				
		SD	0.957	0.489	
Response latencies		mean	57	101	-
3.95	0.001				
		SD	46	52	-

All were free from nervous diseases or injury and had no abnormality in their rest EEG. Informed written consent was obtained from each subject after the

procedures had been fully explained. The local Ethics Committee gave approval for the experiment.

Stimuli. The subjects were presented with one of the following tasks: The length of the diagonal of a square is 5 cm. How long is the length of an edge of a square with double the area? The task requires computation based on a computational or an imagery resolution strategy (Fig. 1). Our subjects had to solve 10 different tasks with this condition.



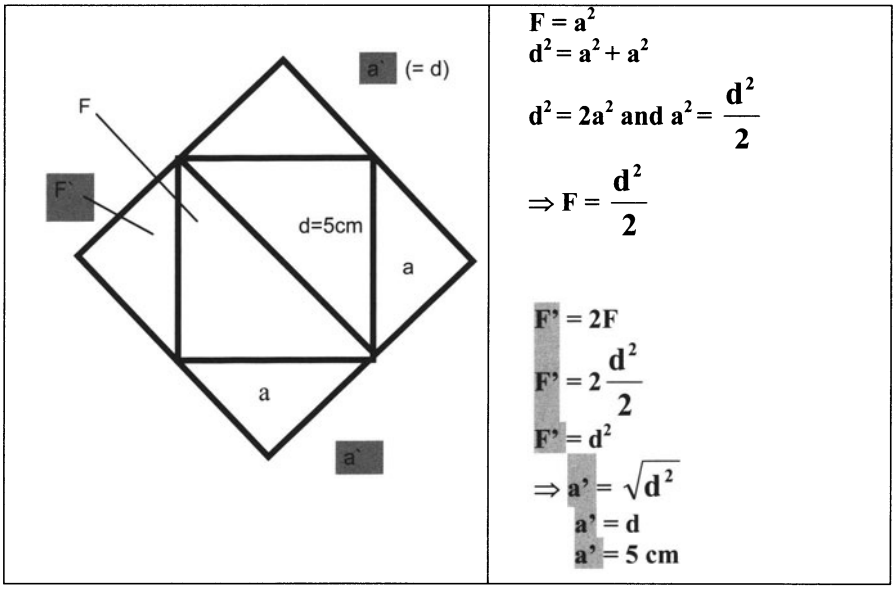


Fig. 1. Example of a mathematical problem with two modality strategies: computational strategy and imagery resolution strategy (Heinrich, 1997)

Procedure. When confronted with tasks with two modality strategies, subjects should activate one of the two possible strategies of computation/imagination . Subjects had to compute the results and give the answer aloud. At the end of each trial, a careful debriefing was carried out individually. Each subject was asked to explain the resolution strategy(ies) used during the solving process.

EEG recording. The EEG was recorded using 19 scalp electrodes. The standard Ag/AgCl electrodes were placed according to the international 10/20 system (Jasper 1958). All cortical electrodes were referenced to two linked ear lobe electrodes during the EEG recording. Data were collected using the Brainstar EEG software linked to a computer for stimulus presentation. The high and low frequency filters were set to 70 and 0.3 Hz, respectively. The data recorded in each channel were digitized at a sampling rate of 256 Hz. The EEG was analyzed in the time interval between the end of text reading and the answer, artefacts were excluded.

Data analysis. Instantaneous coherence analysis (Schack et al. 1995, 1999, 2001) was performed for 30 electrode pairs. The 30-dimensional vector of time courses of band coherences (13-18 Hz[1]) for each single trial was subdivided into segments with stable coherence values. Afterwards, the segments were clustered into six classes. The corresponding cluster centers were designated as microstates. Shannon's entropy with regard to the distribution of the different microstates gives evidence of the disorder of their occurrence. The hidden order may be described by transition probabilities and quantified by the conditional entropy of the occurrence of a microstate by observing the preceeding microstate. The difference between Shannon's entropy and the conditional entropy is denoted by entropy reduction H_{red} :

Shannon's entropy is defined by

$$H(i) = -\sum_{j=1}^{n_0} P(j/i) \cdot \mathrm{ld}(P(j/i)) \tag{1}$$

where j, j = 1, ... , n_0, are the numbers of microstates.

The conditional entropy of occurrence of states under the assumption of i as the present state is defined as

$$H(i) = -\sum_{j=1}^{n_0} P(j/i) \cdot \mathrm{ld}(P(j/i)), \quad i = 1, \ldots, n_0, \tag{2}$$

where ld denotes the logarithm dualis. The difference

$$H_{red} = H - \sum_{i=1}^{n_0} P(i) \cdot H(i) \tag{3}$$

[1] This frequency band was sensitive in earlier experiments (Krause et al. 1998; Schack et al. 1999).

is called entropy reduction and reflects the sequential structure of the states. This indicates a strong sequential structure or concatenation of the states. The depth of the higher order sequential effect was calculated by means of a configuration frequency analysis (Lautsch and Weber 1995). Equation (3) was extended to higher order dependency up to five steps back and tested for inequality. To avoid alpha-inflation the Bonferroni-Holm correction was successfully performed on the level of 5 %.

Results

Behavioral data. Response latencies (RLs) for each sample are given in Table 1. Response latencies were significantly longer for the normal subjects as compared to the gifted subjects in problem solving. Moreover, the samples differ in their percentage of errors (gifted subjects: 15%, SD = 12%; normal subjects: 49%, SD = 23%; $Z = -3.49$, $p = 0.001$). Additionally we obtain results of a post-experimental debriefing.

At the first glance, the task-dependent response latency differences might be explained by the task-dependent different type of resolution. According to the double representation hypothesis, both types of resolution (the computational and the imagery resolution strategy) are accessed by gifted subjects. As a consequence, gifted subjects are able to select the most efficient strategy and to apply it. The double representation hypothesis also postulates that only one resolution strategy is preferably stored in the memory and used by normal subjects. This particular strategy, used by most normal subjects, is less efficient, leading to a longer response latency (Fig. 1, Table 1).

In contrast to the double representation hypothesis, we found in the post-experimental debriefing, that all subjects indicated familiarity with both modality strategies (e.g. computational and visual mental imagery resolution strategy). They reported they had learned application of both modality strategies at school; and had used them in the experiment. Against this background it is worth recalling that *behavioral data can not explain the difference between gifted and normal subjects* and does not answer the question of better cognitive process. Therefore we identify the mental representation and the mental process by means of instantaneous EEG-coherence.

EEG data

Sequential properties of microstates. Essentially, cognitive processes can be described by the microstate sequences derived from the segmentation and clustering of EEG-coherence (Schack et al. 1995; Krause 2000; Schack et al. 2001; Krause et al. 2001). Microstates are EEG-coherence maps which are stable over time. We examine dependencies of microstates in their temporal regime to investigate the difference in performance between gifted and normal subjects.

Entropy reduction H_{red} is a quantitative measure of the strength of concatenation of microstates according to equation (3). Accordingly, we looked for a difference in the entropy reduction between gifted and normal subjects. The gifted subjects exhibit a mean entropy reduction $H_{red} = 1.2$ (SD = 0.13) and thus achieve a significantly better result than their normal counterparts, whose entropy reduction H_{red} was 0.9 (SD = 0.19). The Z value for this comparison is Z = -2.99 ($p < 0.01$). The entropy reduction (Fig. 2) is calculated based on the concatenation of microstates with self-transition.

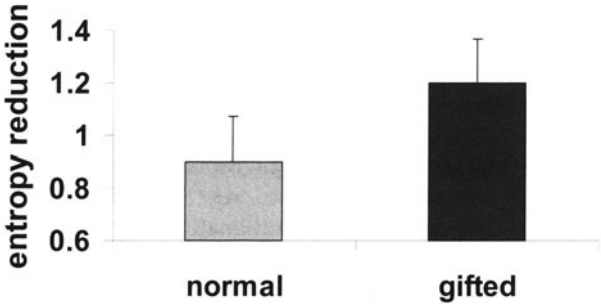


Fig. 2. Response latency and entropy reduction for gifted versus normal subjects (with self-transition of microstates)

Further support for our interpretation of the trend described above can be derived from the calculation of entropy reduction without the self-transition of the microstates. If we exclude the self-transition, the entropy reduction is reduced to concatenation of different microstates. In this case the gifted subjects ($H_{red} = 0.61$, SD = 0.09) exhibit a significantly higher entropy reduction than the normal ($H_{red} = 0.48$, SD = 0.06) subjects (Z = -3.31, $p < 0.01$). The sequence of microstates may be considered as a Markovian chain. The concatenation of microstates without self-transition of the microstates for a gifted and a normal subject is shown in Fig. 3.

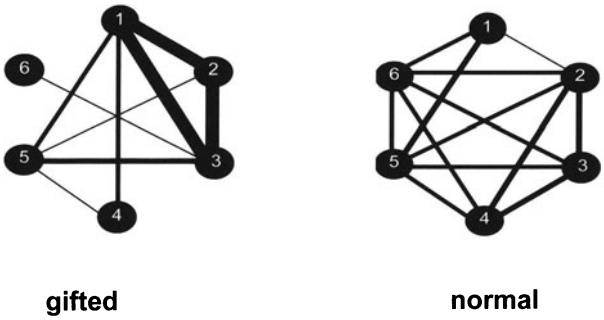


Fig. 3. Sequence of microstates (without self-transition) for one gifted versus one normal subject. The 6 microstates are marked by black circles with numbers. The strength of concatenation is coded by the thickness of the edges

As seen in Fig. 3, the gifted subject exhibits a stronger concatenation of subsequences of microstates.

Higher order sequential effects can be considered in microstate sequences as well. Based on a configuration frequency analysis (Lautsch & Weber 1995) we determine the depth of the sequential effects both for gifted and normal subjects in mathematical problem solving. The result is shown in Fig. 4. The difference of the mean maximum depth of the sequential effect between gifted and normal subjects is significant ($Z = 2.74$, $p = 0.006$). This corroborates the assumption that sequential properties of microstate sequences can differentiate the mathematically gifted from the normal subjects.

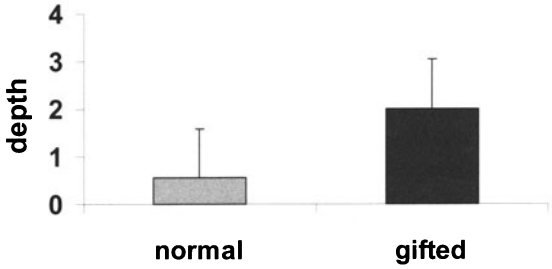


Fig. 4. Mean maximum depth of microstate-sequences (without self-transition) for gifted and normal subjects

Topographical properties of microstates. As mentioned above the two modality strategies in our experiment (computational component of numerical mental calculation and imagery component of figural mental calculation) represent two basic resolution strategies that could be subserved by neural networks belonging to different domains (verbal versus visuospatial). Zargo et al. (2001) give support the idea that numerical and spatial representations are intricately intertwined in the parietal and frontal lobes and that these numerical and visuospatial representations are involved at different levels of mental calculation in normal subjects. In our experiment we find that for gifted subjects the same cortical areas are active: The six microstates (Schack et al. 1999; Seidel 2000) were ranked by means of their self-transition probability for subject groups. For data reduction we restrict our analysis to those microstates with high self-transition. Based on this restriction, Fig. 5 shows the difference coherence maps (gifted minus normal subjects).

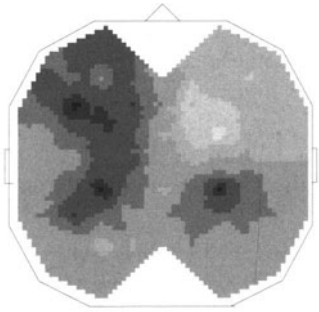


Fig. 5. Difference coherence maps (gifted minus normal subjects) of microstates with high self-transition. The coherence values are scaled between – 0.10 (white colour) and + 0.10 (black colour)

As seen in Fig. 5 gifted subjects exhibit a significantly higher coherence at the electrode pairs F7/F3 ($Z = -2.23$, $p < 0.05$), C3/P3 ($Z = -1.86$, $p < 0.1$) and C4/P4 ($Z = -1.93$, $p < 0.1$). In an unusual comparison of two different types of neuroscientific methods, namely PET – and EEG-studies, this result is in agreement with neural correlates of simple and complex mental calculation by Zago et al. (2001). Zargo et al. found that computing specifically involved two functional networks: a left parieto-frontal network responsible for the holding of the multidigit numbers in visuospatial working memory and a bilateral inferior temporal gyri related to the visual mental imagery resolution strategy. No additional network for mental calculation in gifted subjects is observed in our EEG-study.

Discussion

The main result is focused on the difference of the concatenation of microstates, represented by the sequential properties of the microstate sequence. By solving tasks with two modality strategies, our gifted subjects exhibit a higher entropy reduction and a higher order sequential effect than the normal subjects. The results indicate the gifted subjects' superiority over their normal counterpart in our mathematical problem solving experiments; this is clearly linked to a more deterministic concatenation of different microstates. This higher concatenation of microstates in the case of the gifted subjects might be interpreted as a better context-dependent application of the most efficient modality strategy. The result does not exclude, however, the possibility that in special cases other components may also contribute to sequential dependencies.

The difference between mathematically gifted and normal subjects appears in the topographical properties of the microstates as well. The activation of the computational network in gifted subjects is distinctly increased when tasks were processed with two modality strategies, suggesting that gifted subjects process tasks with two modality strategies better than normal subjects. Looking for the reason for this different activation, one might speculate that gifted subjects, in contrast to their normal counterparts, are able to better access both modality strategies, although their debriefing reports do not differ. This finding corroborates the double representation hypothesis and stresses the sensitivity of this method.

One could argue that the higher coherence at the electrode pairs C3/P3 and C4/P4 (Fig. 5) observed for gifted subjects might be linked to a better visualization and is not a feature of the double representation hypothesis in giftedness. Evidence for this assumption comes from the study by Jaencke et al. (1998) in which mental rotation tasks were presented and the fMRI activation was registered. Subjects with higher score values in the mental rotation test achieved a higher activation in the left and right parietal cortical areas and vice versa. The score values in the mental rotation test of our gifted subjects and their normal

counterparts do not differ (Table 1). Therefore we reject the visualization hypothesis.

The present study has introduced a new method of investigating problem solving using microstate sequences based on EEG-coherence. As an alternative to the IQ testing, this method may be extended to measure intelligence more objectively.

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References

Heinrich F (1997) Diskussionsmaterial zur Untersuchung der Doppelrepräsentations-hypothese und ein Bemerkungen aus mathematikdidaktischer Sicht. (unveröffentlicht).

Jäncke L, Kleinschmidt A, Mirzazade S, Specht K, Freund HJ (1998) Mental rotation ability determines the activity in posterior cortical areas during the imagination, palpation and construction of 3D objects. NeuroImage **7**: 119

Jasper HH (1958) The ten-twenty electrode system of the International Federation. Electroencephalography and Clinical Neurophysiology 10: 371-375

Klix F (1992) Die Natur des Verstandes. Hogrefe, Göttingen

Krause W (2000) Denken und Gedächtnis aus naturwissenschaftlicher Sicht. Hogrefe, Göttingen

Krause W, Seidel G, Schack B (2001) Ordnungsbildung. Z. Psychol. 209: 376-401

Lautsch E, Weber S (1995) Methoden und Anwendungen der Konfigurationsfrequenzanalyse (KFA). Belz Psychologie Verlags Union, Weinheim

Schack B, Krause W (1995) Dynamic power and coherence analysis of ultra short-term cognitive processes--a methodical study. Brain Topogr 8: 127-136

Schack B, Grieszbach G, Krause W (1999) The sensitivity of instantaneous coherence for considering elementary comparison processing. Part I: the relationship between mental activities and instantaneous EEG coherence. Int. J. Psychophysiol. 31: 219-240

Schack B, Seidel G, Krause W, Heinrich F (2001) Coherence Analysis of the Ongoing EEG by Means of Microstates of Synchronous Oscillations. Proceedings-23rd Annual Conference-IEEE/EMBS Oct 25-28, Istanbul

Seidel G. (2000) Ordnungsbildung und Doppelrepräsentation im Denken mathematisch , Hochbegabter. Sequentielle und topologische Eigenschaften von Mikrozustandssequenzen. Doctoral thesis, University of Jena.

Zargo L, Pesenti M, Mellet E, Crivello F, Mazoyer B, Tourio-Mazoyer N (2001) Neural Correlates of Simple and Complex Mental Calculation. NeuroImage 13: 314-327